



OPEN ISRSL0: compressed sensing MRI with image smoothness regularized-smoothed ℓ_0 norm

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The reconstruction of MR images has always been a challenging inverse problem in medical imaging. Acceleration of MR scanning is of great importance for clinical research and cutting-edge applications. One of the primary efforts to achieve this is using compressed sensing (CS) theory. The CS aims to reconstruct MR images using a small number of sampled data in k-space. The CS-MRI techniques face challenges, including the potential loss of fine structure and increased computational complexity. We introduce a novel framework based on a regularized sparse recovery problem and a sharpening step to improve the CS-MRI approaches regarding fine structure loss under high acceleration factors. This problem is solved via the Half Quadratic Splitting (HQS) approach. The inverse problem for reconstructing MR images is converted into two distinct sub-problems, each of which can be solved separately. One key feature of the proposed approach is the replacement of one sub-problem with a denoiser. This regularization assists the optimization of the Smoothed ℓ_0 (SL0) norm in escaping local minimums and enhances its precision. The proposed method consists of smoothing, feature modification, and Smoothed ℓ_0 cost function optimization. The proposed approach improves the SL0 algorithm for MRI reconstruction without complicating it. The convergence of the proposed approach is illustrated analytically. The experimental results show an acceptable performance of the proposed method compared to the network-based approaches.

Keywords MRI reconstruction, Sparse, Compressed sensing, K-space, Smoothed norm

Over the past few decades, Magnetic Resonance Imaging (MRI) has become a leading method for diagnosing and investigating various diseases. Additionally, its avoidance of harmful rays like X-rays helps minimize damage to the body¹. MRI employs radio frequency waves to non-invasively evaluate soft tissues' structure, function, and morphology². However, unlike other clinical imaging methods like computed tomography (CT) or ultrasound, MRI faces the challenge of slow data acquisition. Fast MRI techniques are crucial to accelerate data acquisition while maintaining high-quality image reconstruction. In recent decades, compressed sensing (CS) has emerged as a popular and successful strategy for rapidly reconstructing MR images. Compressed Sensing MRI (CS-MRI), as an effective approach, enables a much lower data sampling rate than the Nyquist rate without significant degradation in image quality^{3–5}.

Compressed sensing exploits the fact that an image is usually sparse in some appropriate transform domains^{6,7}, provided that the acquisition process adheres to principles of incoherence. In this context, incoherence is crucial for disrupting the conventional regularity in sampling patterns, enabling reconstruction methods based on sparsity⁸. The concept of “incoherent” refers to the deliberate design of sampling patterns when acquiring MRI data. This term indicates a deliberate lack of correlation or regularity in how the data is sampled. The goal is to deviate from the typical systematic or predictable sampling patterns. It facilitates reconstructing high-quality images from a reduced set of non-redundant samples. Within most early CS-MRI methodologies, the model consists of two main components: a data fidelity term in the k-space domain and a regularization term of sparse representation in different domains, such as finite-difference⁹ and wavelet transforms¹⁰. However, a key disadvantage of these classical methods is the noticeable loss of structures and details in the reconstructed images, especially when dealing with high under-sampling rates.

Since the introduction of sparse image representation, a significant focus has been on reconstructing MR images using CS. Numerous approaches have been developed to address the inverse problem within this framework¹¹. One such method is the Fast Multiclass Dictionary Learning (FDLCP) algorithm¹², which optimally determines a dictionary matrix for image reconstruction. FDLCP's advantage is its unique approach to multiclass dictionary learning, enabling efficient reconstruction from under-sampled data. However, it struggles with

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computational complexity and may not generalize well to different types of MR images. Iterative thresholding techniques, including the Smoothing Fast Iterative Shrinkage-Thresholding Algorithm (SFISTA)¹³ and Projected Fast ISTA (PFISTA)¹⁴, have also significantly contributed to solving the inverse problem, particularly in the context of CS-MRI.

In recent years, deep neural networks have achieved notable achievements in various domains, such as image classification¹⁵, segmentation¹⁶, denoising¹⁷, and high-speed MRI. The primary advantage of deep learning methods is their ability to automatically learn complex features and patterns from data, making them highly effective for MRI reconstruction. For instance, the Iterative Feature Refinement Network (IFR-Net)¹⁸, operating within the Iterative Feature Refinement Compressed Sensing (IFR-CS) model¹⁹, integrates insights from parameter learning within the Alternating Directions Method of Multipliers Net (ADMM-Net)²⁰ and learning convolutional neural network (CNN) weights. This approach is advantageous due to its ability to refine features iteratively, leading to high-quality reconstructions. However, the need for large amounts of training data and the high computational cost are significant disadvantages. In²¹, Wu et al. presented the K-space Transformer MRI Reconstruction Model (KTMR), an approach designed to reconstruct magnetic resonance (MR) images from under-sampled k-space data. Drawing upon the Swin Transformer architecture, KTMR provides an effective solution for generating fully sampled MR images across diverse anatomical structures. Combining the advantages of CNNs and Transformer models, their method offers an end-to-end framework that seamlessly transforms under-sampled images into their fully sampled counterparts²².

To achieve the sparsest representation of MR images, the Smoothed ℓ_0 norm (SL0) is a suitable choice²³, quantifying the minimum number of nonzero components in an MR image. The main advantage of SL0 is its computational efficiency, being 2-3 times faster than Basis Pursuit (BP) while maintaining comparable or superior accuracy²⁴. However, it struggles with local minima, particularly when the sparsity level is high, and it can be sensitive to noise²³.

This paper adds an image smoothness-based regularization term to the smoothed norm to recover the MRI image. This regularization is used to avoid local minima trapping in the optimization process. This research makes significant contributions to the field of MRI reconstruction through the following innovations:

- We propose a novel modification to the classic SL0 algorithm by incorporating an image smoothness-based regularization term. This addition significantly improves the algorithm's ability to avoid local minima, leading to more accurate and reliable reconstructions.
- The proposed algorithm demonstrates powerful performance across various sampling masks and different sampling rates, showcasing its versatility and robustness in handling diverse MRI acquisition scenarios. This adaptability further underscores the algorithm's potential for practical clinical applications where varying sampling conditions are common.
- The proposed optimization problem is effectively solved using the HQS method, which decomposes the problem into two subproblems: denoising and sparse recovery. This approach ensures efficient and stable convergence.
- The proposed algorithm demonstrates competitive performance when compared to state-of-the-art deep learning methods, with the added advantages of simplicity and low computational complexity. The experimental results underscore its superiority over classical methods and its potential for real-time clinical applications.

This research advances the current state of MRI reconstruction by offering a solution that balances computational efficiency with high reconstruction quality, making it a valuable tool for both clinical and research applications.

In the following sections, we describe the proposed algorithm in “Section [Proposed method](#)”, explaining its mathematical foundations. “Section [Convergence Analysis](#)” evaluates the analysis of the algorithm's convergence, while “Section [Experimental result](#)” provides a detailed examination of the empirical results of the proposed approach applied to brain and knee datasets. Finally, “Section [Discussion & conclusion](#)” provides the concluding remarks of the research.

Proposed method

Various methods have been developed to obtain the sparse solution. The SL0 method has emerged as a fast and accurate approach. This study aims to improve the SL0 algorithm, enhancing its precision while maintaining low complexity. In pursuit of this objective, we integrate a smoothness condition into the algorithmic framework to minimize the risk of converging to local minimums. However, this smoothness may unintentionally remove structural elements for analysis. This side effect is controlled by two techniques: a decreasing factor reduces the smoothness level in each iteration, and a “sharpening” operator is used to counterbalance the loss. This approach ensures that delicate structures, especially edges crucial for MR image analysis, are returned to the original image during each iteration. In this section, the convergence analysis is also presented.

In the rest of this section, the SL0 idea is stated first, then the main idea and the optimization process are presented.

Smoothed ℓ_0 norm

The SL0 norm optimization technique is a popular approach in sparse signal/ image recovery, where the goal is to approximate the ℓ_0 norm by a smooth, continuous function that is easier to optimize. The SL0 method was originally proposed as an alternative to the traditional ℓ_1 norm minimization techniques, which are widely used in compressed sensing.

In SL0 optimization, the ℓ_0 norm is approximated by a family of smooth functions that converge to the ℓ_0 norm as a smoothing parameter tends to zero. This approach allows for efficient optimization, as it transforms

the non-convex ℓ_0 problem into a series of smoother problems that can be solved using standard optimization techniques. The SL0 method is particularly effective in applications where sparse solutions are desired, such as image reconstruction, compressed sensing, and sparse coding. This function is minimized using the steepest descent algorithm. The core of this approximation is a continuous function serving as an estimator for the ℓ_0 norm of $\mathbf{s} = [s_i] \in \mathbb{R}^m$, denoted as $\|\mathbf{s}\|_0$, that counts the nonzero elements of the vector $\mathbf{s}$. Consider the following smoothed ℓ_0 norm function.

$$\|\mathbf{s}\|_{0\sigma} = m - F_\sigma(\mathbf{s}) = m - \sum_{i=1}^m \exp\left(\frac{-s_i^2}{2\sigma^2}\right) \approx \|\mathbf{s}\|_0 \quad (1)$$

Where $F_\sigma(\mathbf{s}) = \sum_{i=1}^m \exp\left(\frac{-s_i^2}{2\sigma^2}\right)$. This function includes a crucial parameter, σ , which significantly influences the quality of the approximation. The value of σ determines the trade-off between the smoothness of the function and the accuracy of the smooth ℓ_0 norm approximation. For small values of σ , $\|\mathbf{s}\|_0$ can be approximated as $m - F_\sigma(\mathbf{s})$ and this approximation approaches equality as σ approaches 0. Thus, finding the minimum ℓ_0 norm solution involves maximizing $F_\sigma(\mathbf{s})$ (subject to $\mathbf{A}\mathbf{s} = \mathbf{y}$) with a very small σ . It's important to note that the σ value affects the function's smoothness: Larger values of σ result in a smoother function with a less accurate ℓ_0 norm approximation and fewer local maxima, while smaller σ values provide a closer approximation to the ℓ_0 norm but yield a highly non-smooth function with many local maxima, making the maximization more challenging. To consider this problem, the algorithm employs a decreasing sequence of σ values and a gradient descent approach. By gradually reducing σ , each iteration of maximization begins with an initial solution that is close to the prior $F_\sigma(\mathbf{s})$ maximizer. This strategy ensures that a slight change in σ results in the new maximizer being in proximity to the previous maximizer of $F_\sigma(\mathbf{s})$ ²³. Although this approach reduces concerns related to local maxima, it does not entirely resolve the issue. Figure 1 illustrates the SL0 function used to approximate the ℓ_0 norm, demonstrating how the smooth approximation allows for more efficient optimization while retaining the characteristics of the original norm.

To enhance understanding of the proposed algorithm steps and how smoothness regularization aids in the convergence of the SL0 algorithm, we provide a simplified explanation of each component:

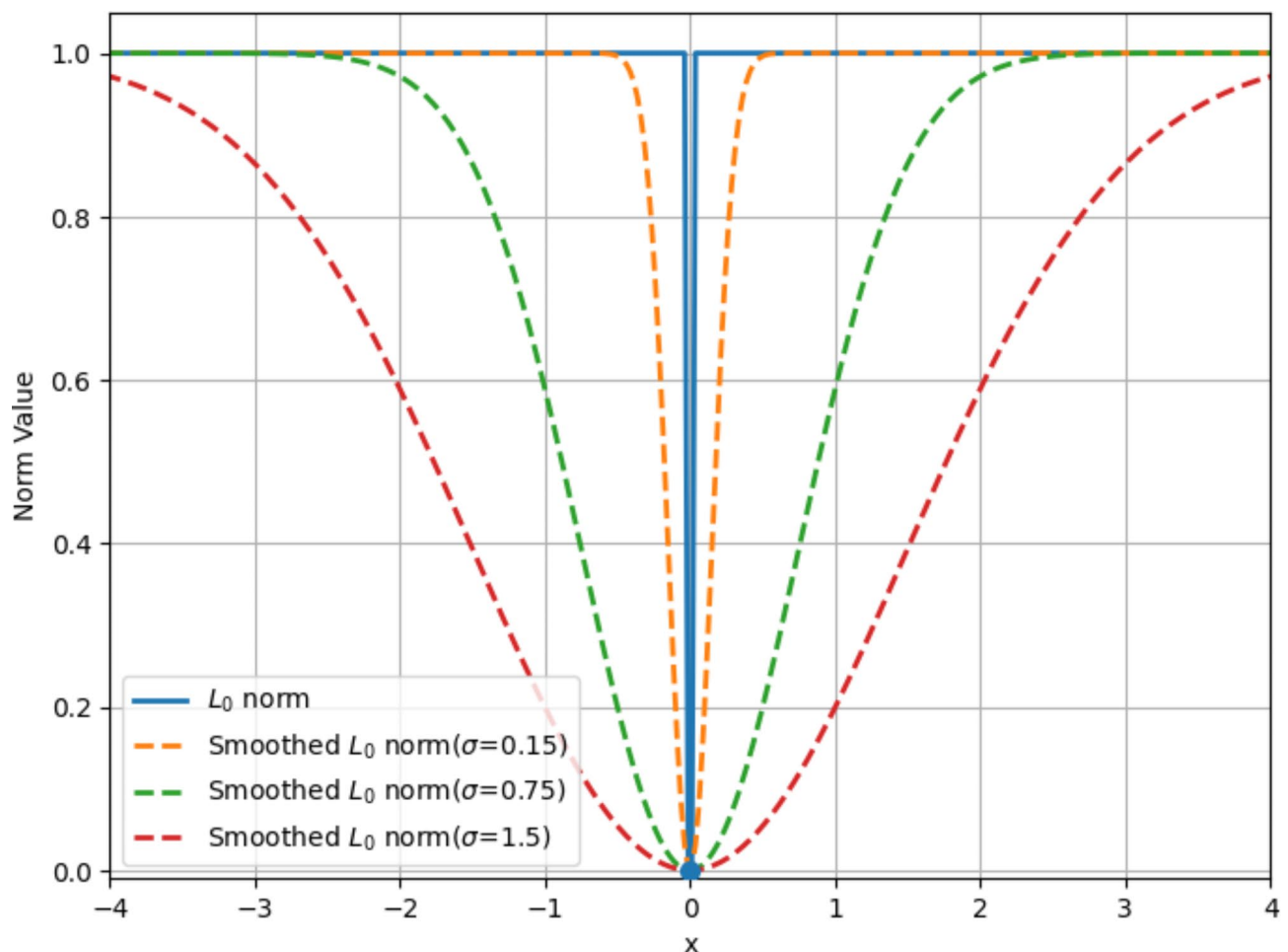


Fig. 1. The function of smoothed ℓ_0 norm.

- The **sparse representation** problem aims to find the **most compact data representation**, **minimizing non-zero elements**. However, solving this problem exactly is NP-hard due to the discrete nature of the ℓ_0 norm. To overcome this, methods like the **SL0 approximate the ℓ_0 norm with a continuous function**, making optimization more feasible while maintaining sparsity.
- The SL0 algorithm approximates the ℓ_0 norm using a smooth, differentiable function, enabling efficient gradient-based optimization techniques. **It gradually reduces the sigma (σ) parameter to sharpen this approximation over iterations**, **preventing premature convergence to local minima** and improving the focus on sparsity.
- Despite smoothing, the **SL0 function is still non-convex**, leading to potential traps in local minima, especially with sparse data or limited measurements. This non-convexity can hinder the algorithm from achieving the optimal sparse solution.
- **Role of Smoothness Regularization:** The natural smoothness in images, such as MRI scans, can be leveraged to improve the SL0 algorithm's convergence. By adding a smoothness regularization term, the optimization process is more effectively guided, **reducing overfitting and the risk of local minima**. **As iterations progress, the influence of the smoothness term lessens, allowing the algorithm to better focus on sparsity**. This **balance between smoothness and sparsity** enhances convergence, leading to more accurate and robust image reconstructions.

This combination of techniques demonstrates how the SL0 algorithm, with smoothness regularization, can effectively navigate the challenges of sparse representation in image processing, ultimately improving performance in practical applications.

Main idea

In this section, a regularization is added to the sparse recovery problem to improve the performance of the SL0 algorithm. Consider a discrete signal $\mathbf{s} \in \mathbb{R}^n$ which has a sparse representation $\boldsymbol{\alpha}$ in a transform domain $\boldsymbol{\Psi} \in \mathbb{R}^{n \times n}$ such as wavelet, Fourier²⁵, or DCT^{26,27}. The signal $\mathbf{s}$ is measured by a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ and expressed as follows.

$$\mathbf{y} = \mathbf{A}\mathbf{s} = \mathbf{A}\boldsymbol{\Psi}\boldsymbol{\alpha} \quad (2)$$

In the application of image reconstruction, image recovery is converted to sparse recovery, i.e., finding the sparsest solution of the linear model (2). Here, the smoothed ℓ_0 norm is used as a sparsity measure. The optimization problem can be formulated as:

$$\hat{\boldsymbol{\alpha}} = \min \|\boldsymbol{\alpha}\|_{0\sigma} \quad \text{s.t.} \quad \mathbf{y} = \mathbf{A}\boldsymbol{\Psi}\boldsymbol{\alpha} \quad (3)$$

Since natural images such as MRI are smooth with minimum total variation, this knowledge improves image recovery by combining the SL0 approach and smoothness. The SL0 tries to optimize the ℓ_0 norm directly by gradually decreasing the parameter σ ; hence, there is a challenge of trapping in a local minimum in this approach. This combination can effectively consider this issue. Therefore, we propose the following Image Smoothness-regularized SL0 (ISRSL0) problem.

$$\text{ISRSL0:} \quad \hat{\mathbf{s}} = \underset{\mathbf{s}}{\operatorname{argmin}} \|\boldsymbol{\Psi}^{-1}\mathbf{s}\|_{0\sigma} + \lambda_s \text{smoothness}(\mathbf{s}) \quad \text{s.t.} \quad \mathbf{y} = \mathbf{A}\mathbf{s} \quad (4)$$

where λ_s is the regularization parameter that controls the balance between smoothness and sparseness, we recast the ISRSL0 problem in the following mathematical form

$$\text{ISRSL0:} \quad \hat{\mathbf{s}} = \underset{\mathbf{s}}{\operatorname{argmin}} J(\mathbf{s}), \quad J_{\text{ISRSL0}}(\mathbf{s}) = \|\boldsymbol{\Psi}^{-1}\mathbf{s}\|_{0\sigma} + \lambda_s \text{smoothness}(\mathbf{s}) + \delta_c(\mathbf{s}) \quad (5)$$

where $\delta_c(\mathbf{s})$ represents the subspace of measurement data $C = \{\mathbf{s} \in \mathbb{R}^n | \mathbf{y} = \mathbf{A}\mathbf{s}\}$, defined as

$$\delta_C(\mathbf{s}) = \begin{cases} 0 & \text{if } \mathbf{s} \in C \\ +\infty & \text{if } \mathbf{s} \notin C \end{cases} \quad (6)$$

The smoothness regularization can be any measure function such as total variation or deep image prior. We don't define this cost function implicitly because the proposed optimization approach considers its effect in a denoising step.

The HQS approach²⁸ optimizes the ISRSL0 cost function $J(\mathbf{s})$. The HQS method is a well-known technique used in different applications of inverse problems. It divides the minimization function into two components: one related to the model space and a regularization term achieved by introducing an auxiliary variable.

In the HQS method, an auxiliary variable $\mathbf{z} \in \mathbb{R}^n$ is introduced to formulate the minimization problem as follows:

$$\hat{\mathbf{s}}, \hat{\mathbf{z}} = \underset{\mathbf{s}}{\operatorname{argmin}} J_{\text{HQS}}(\mathbf{s}), \quad \text{s.t.} \quad \mathbf{z} = \mathbf{s} J_{\text{HQS}}(\mathbf{s}, \mathbf{z}) = \|\boldsymbol{\Psi}^{-1}\mathbf{s}\|_{0\sigma} + \lambda_s \text{smoothness}(\mathbf{z}) + \delta_c(\mathbf{s}) \quad (7)$$

To solve this problem, the penalty method is employed²⁹, resulting in the following expression:

$$\hat{\mathbf{s}}, \hat{\mathbf{z}} = \underset{\mathbf{s}}{\operatorname{argmin}} J_{\text{HQS}}^p(\mathbf{s}), \quad \text{s.t.} \quad \mathbf{z} = \mathbf{s} J_{\text{HQS}}^p(\mathbf{s}, \mathbf{z}) = \|\boldsymbol{\Psi}^{-1}\mathbf{s}\|_{0\sigma} + \lambda_s \text{smoothness}(\mathbf{z}) + \delta_c(\mathbf{s}) + \frac{1}{2\lambda_p} \mathbf{s} - \mathbf{z}_2^2 \quad (8)$$

Where λ_p is the penalty parameter. This parameter controls deviation from the condition $\mathbf{z} = \mathbf{s}$. The HQS approach converts the minimization problem (8) into two disjoint subproblems as follows:

$$\begin{cases} \mathbf{z}_{k+1} = \underset{\mathbf{z}}{\operatorname{argmin}} J_{HQS}^p(\mathbf{s}_k, \mathbf{z}) \\ \mathbf{s}_{k+1} = \underset{\mathbf{s}}{\operatorname{argmin}} J_{HQS}^p(\mathbf{s}, \mathbf{z}_{k+1}) \end{cases} \quad (9)$$

These steps can be simplified into the following mathematical form, each term with an explicit concept.

$$\begin{cases} \mathbf{z}_{k+1} = \underset{\mathbf{z}}{\operatorname{argmin}} \lambda_s \text{smoothness}(\mathbf{z}) + \frac{1}{2\lambda_p} \|\mathbf{s}_k - \mathbf{z}\|_2^2 \\ \mathbf{s}_{k+1} = \underset{\mathbf{s}}{\operatorname{argmin}} \|\Psi^{-1}\mathbf{s}\|_{0\sigma} + \frac{1}{2\lambda_p} \|\mathbf{s} - \mathbf{z}_{k+1}\|_2^2 + \delta_c(\mathbf{s}) \end{cases} \quad (10)$$

The first subproblem is dedicated to implementing the smoothing condition, while the second reconstructs the image through SL0 optimization within subspace C . The following subsections investigate these optimization problems.

First subproblem: denoising and sharpening

A more precise evaluation reveals that the first subproblem of (10) can be expressed as the denoising of $\mathbf{s}_k$. In this subproblem, $\frac{1}{2\lambda_p} \|\mathbf{s}_k - \mathbf{z}\|_2^2$ has the role of the fidelity term, and the regularization term is any smoothness criteria for denoising. So, the optimization solution $\mathbf{z}_{k+1}$ is defined as follows:

$$\mathbf{z}_{k+1} = D(\mathbf{s}_k, g(k)) \quad (11)$$

where $D(\cdot, g(k))$ represents any denoiser such as BM3D³⁰, Wiener filter²⁷, or deep learning models. The parameter $g(k)$ denotes the noise level and controls the smoothness level. The proposed parameter $g(k)$ is in the following form.

$$g(k) = \frac{0.08}{\text{Sampling Rate}\%} \times \sigma_k \quad (12)$$

This parameter is a decreasing sequence. The small $g(k)$ is equivalent to decreasing the effect of the smoothness regularization in the optimization problem, eliminating less essential image details and structures. The proposed value of the parameter $g(k)$ is proportional to the parameter σ in the SL0 algorithm. As the iterations progress, the smoothing effect gradually diminishes, leaving only the SL0 term in the final stages. In the sparse recovery problem, decreasing the sampling rate or the ratio $\frac{m}{n}$ highlights the challenge of local minima. To consider this issue in the proposed algorithm, the denoiser parameter is inversely related to the sampling rate. Our experimental results demonstrate that a smoothing coefficient of 0.08 has a minimal impact on reconstruction quality, with variations within the range of 0.01 to 0.15, leading to a PSNR standard deviation of 0.1dB. This specific range of smoothing coefficients was meticulously chosen to balance the compromise between the intensity of smoothing applied and the preservation of image details. It would enhance the clarity and accuracy of the final output.

However, due to the inherent smoothing effect, the denoising process can result in a loss of relevant data, including edges and fine details. To rectify this issue and prevent the permanent removal of such information by the denoiser, the “sharpening” operator is employed. This operator is pivotal in restoring critical image components, ensuring that only the fundamental elements are reintegrated into the final image.

The main goal is to preserve essential image structures that are inadvertently removed during denoising. The Sharpening function is employed to achieve this, which evaluates local contrast, brightness, and correlation between pre-smoothing and post-smoothing images³¹. This modifier facilitates the recovery of valuable components and acts like a fine sieve in image processing. Over iterations, a diminishing coefficient gradually reduces the sharpening effect.

The process starts with a Gaussian filter applied to the denoised image $\mathbf{z}_{k+1}$. The convolution, denoted as $G_{\sigma_t} * \mathbf{z}_{k+1}$, filters the image using a linear Gaussian filter with σ_t as the standard deviation. p and q represent two local image patches centered around pixel x , derived from the denoised and degraded images $\mathbf{z}_{k+1}$ and $\mathbf{s}_k$. The Sharpening function, SR, is computed as follows:

$$SR = 1 - \left| \frac{3\sigma_{pq} + v}{2\sigma_p^2 + 2\sigma_q^2 + v} \right| \quad (13)$$

The inclusion of v in (13) for numerical stability. The SR adds some removed details and structures to the denoised image $\mathbf{z}_{k+1}$. Figure 2 presents the block diagram of the proposed approach.

Second subproblem: gradient descent and projection

The second optimization step in (10) is a regularized version of the SL0 algorithm. Inspired by the SL0 optimization, this problem is iteratively solved in that each iteration contains the following two steps: gradient descent and projection.

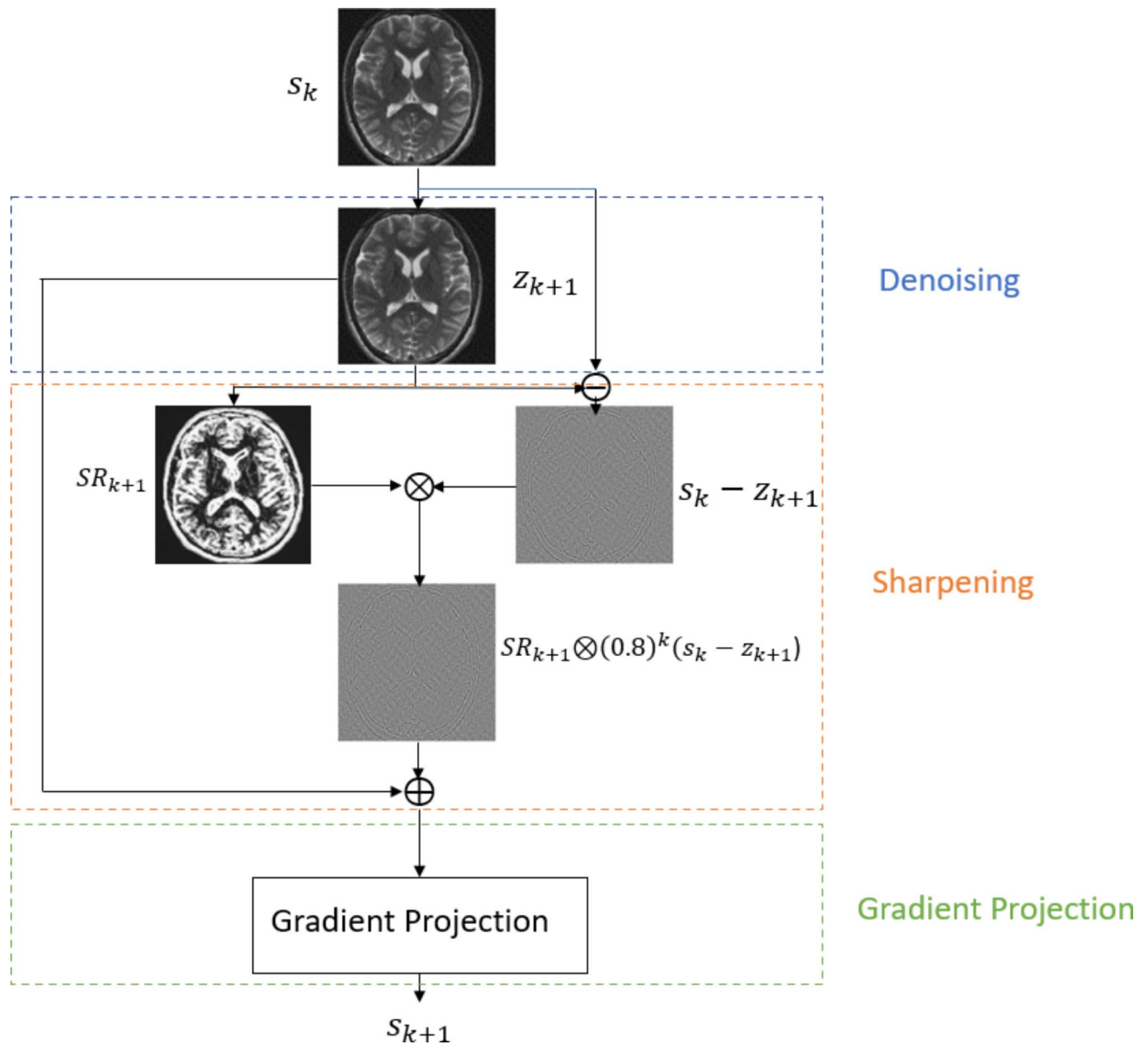


Fig. 2. Block diagram of the proposed algorithm.

$$\begin{cases} \mathbf{s} = \arg \min_{\mathbf{s}} \left\| \Psi^{-1} \mathbf{s} \right\|_{0\sigma} + \frac{1}{2\lambda_p} \left\| \mathbf{s} - \mathbf{z}_{k+1} \right\|_2^2 \\ \text{Project } \mathbf{s} \text{ onto the feasible set } C \end{cases} \quad (14)$$

In the optimization of the cost function $\left\| \Psi^{-1} \mathbf{s} \right\|_{0\sigma} + \frac{1}{2\lambda_p} \left\| \mathbf{s} - \mathbf{z}_{k+1} \right\|_2^2$, one iteration gradient descent is used. In the proposed approach, the wavelet transform $\Psi \in R^{n \times n}$ is used to obtain a sparse representation of the image. Since this transform is unitary, with simple computation, we obtain:

$$\left\| \mathbf{s} - \mathbf{z}_{k+1} \right\|_2^2 = \left\| \Psi \alpha - \Psi \beta_{k+1} \right\|_2^2 = \left\| \alpha - \beta_{k+1} \right\|_2^2 \quad (15)$$

where $\alpha = [\alpha_i] \in R^n$ and $\beta_{k+1} \in R^n$ are the wavelet coefficients of $\mathbf{s}$ and $\mathbf{z}_{k+1}$, respectively. To compute the gradient of this cost function, based on this equality, the cost function $\left\| \Psi^{-1} \mathbf{s} \right\|_{0\sigma} + \frac{1}{2\lambda_p} \left\| \mathbf{s} - \mathbf{z}_{k+1} \right\|_2^2$ is recast in the form of $\left\| \alpha \right\|_{0\sigma} + \frac{1}{2\lambda_p} \left\| \alpha - \beta_{k+1} \right\|_2^2$. The gradient descent iteration can be computed in the following mathematical form:

$$\mathbf{s} \leftarrow \Psi \alpha - \mu \Psi \left(\sum_{i=1}^M \exp \left(\frac{-\alpha_i^2}{2\sigma^2} \right) + \frac{1}{\lambda_p} (\alpha - \beta_{k+1}) \right) \quad (16)$$

The projection process is defined as applying Euclidean mapping and projecting $\mathbf{s}$ onto subspace C . This step is represented as:

$$\mathbf{s} \leftarrow \mathbf{s} - \mathbf{A}^\dagger (\mathbf{A}\mathbf{s} - \mathbf{y}) \quad (17)$$

where $\mathbf{A}^\dagger = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}$ is the pseudo inverse of $\mathbf{A}$.

Matrix computation is needed for the above computation. In the MRI application, the matrix sizes are large. Hence, the required memory and computational complexity restrict the implementation of the projection step. To address this issue, the operators must implement the projection considering the model data in the MRI imaging. The CS-MRI model can be shown as follows:

$$\mathbf{y} = \mathbf{A}\mathbf{s} = \mathbf{M}\mathbf{F}\mathbf{s} \quad (18)$$

where $\mathbf{y} \in R^m$ represents the under-sampled image in the frequency domain, $\mathbf{M} \in R^{m \times n}$ denotes the sampling or measurement matrix, and $\mathbf{F} \in R^{n \times n}$ represents the Fourier operator. The matrix $\mathbf{F}$ is unitary, i.e., $\mathbf{F}^T\mathbf{F} = \mathbf{F}\mathbf{F}^T = \mathbf{I}_n \in R^{n \times n}$ where $\mathbf{I}_n$ is the identity matrix. The matrix $\mathbf{M}$ has only one nonzero element in each row, and each of two its rows is orthogonal, i.e., $\mathbf{M}\mathbf{M}^T = \mathbf{I}_m \in R^{m \times m}$. Based on these properties, the pseudo inverse of $\mathbf{A}$ can be simplified to:

$$\mathbf{A}^\dagger = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1} = \mathbf{F}^T\mathbf{M}^T(\mathbf{M}\mathbf{F}\mathbf{F}^T\mathbf{M}^T)^{-1} = \mathbf{F}^T\mathbf{M}^T \quad (19)$$

According to the above simplification, the projection step is expressed as:

$$\mathbf{s} \leftarrow \mathbf{s} - \mathbf{F}^T (\mathbf{M}^T\mathbf{M}\mathbf{F}\mathbf{s} - \mathbf{M}^T\mathbf{y}) \quad (20)$$

The measurement matrix $\mathbf{M}$ can be defined as the sampling pattern or Mask. According to (18), the following simplification can be concluded to reduce the computational complexity.

$$\mathbf{M}^T\mathbf{y} = \mathbf{M}^T\mathbf{M}\mathbf{F}\mathbf{s} = \text{Mask} \odot \mathbf{F}\mathbf{s} \quad (21)$$

where **Mask** signifies the sampling pattern, comprising binary values (zero and one), and $\odot$ indicates element-wise multiplication. Based on the mentioned note, the projection can be expressed in the following form.

$$\mathbf{s} \leftarrow \mathbf{s} - \mathbf{F}^T (\text{Mask} \odot \mathbf{F}\mathbf{s} - \hat{\mathbf{y}}) \quad (22)$$

where $\hat{\mathbf{y}} \in R^n$ is introduced as the zero-padded $\mathbf{y}$. This is the Fourier transform of the original image, where the acquired coefficients by the imaging process are known, and the remaining coefficients are set to zero. The projection expression (22) does not need any matrix inversion.

IRSL0 algorithm

The IRSL0 algorithm consists of two loops. The outer loop is designed to decrease the parameter σ gradually to avoid being trapped in a local minimum. The outer loop will continue until the condition $\sigma > \sigma_{min}$ is satisfied. The denoiser and sharpening operations are in this loop. The inner loop employs the Gradient Projection algorithm, encompassing two fundamental components: the gradient descent and projection stage. In the gradient descent step, the objective is to minimize the cost function $\|\Psi^{-1}\mathbf{s}\|_{0\sigma} + \frac{1}{2\lambda_p}\|\mathbf{s} - \mathbf{z}_{k+1}\|_2^2$ for a specified σ value. The inner loop is repeated in several iterations ($L=2$). Convergence within this inner loop isn't a strict requirement. The proposed IRSL0 algorithm to reconstruct an MRI image is illustrated in Algorithm 1.

Algorithm 1: ISRSL0**Input:** observation vector $\mathbf{y}$, sampling pattern **Mask**, penalty parameter λ_p , gradient descent parameter μ **Initialization:**

- $\mathbf{s}_0 = \mathbf{A}^\dagger \mathbf{y} = \mathbf{F}^T \mathbf{M}^T$
- Choose a suitable decreasing sequence, $[\sigma_1, \dots, \sigma_K]$.
- $g(k) = \frac{0.08}{\text{Sampling Rate}\%} \times \sigma_k$

For $k = 1, \dots, K$

- Denoising: $\mathbf{z}_k = D(\mathbf{s}_{k-1}, g(k))$

- Sharpening:

- 1) $SR = 1 - \left| \frac{3\sigma_p q + v}{2\sigma_p^2 + 2\sigma_q^2 + v} \right|$
- 2) $\mathbf{z}_k = \mathbf{z}_k + SR \otimes ((0.8)^k \cdot \mathbf{s}_{k-1})$

- $\mathbf{s} = \mathbf{s}_{k-1}$

- Minimizing $\|\Psi^{-1} \mathbf{s}\|_{0\sigma} + \frac{1}{2\lambda_p} \|\mathbf{s} - \mathbf{z}_{k+1}\|_2^2 + \delta_c(\mathbf{s})$ using L iterations of the gradient descent and projection on the feasible subspace $\mathcal{C}$:

- **For** $j = 1, \dots, L$

- 1) $\boldsymbol{\alpha} = \Psi \mathbf{s}$

- 2) $\boldsymbol{\beta}_k = \Psi \mathbf{z}_k$

- 3) Gradient descent: $\mathbf{s} \leftarrow \Psi \boldsymbol{\alpha} - \mu \Psi \left(\sum_{i=1}^M \exp\left(\frac{-\alpha_i^2}{2\sigma_k^2}\right) + \frac{1}{\lambda_p} (\boldsymbol{\alpha} - \boldsymbol{\beta}_k) \right)$

- 4) Projection into the feasible subspace $\mathcal{C}$

$$\mathbf{s} \leftarrow \mathbf{s} - \mathbf{F}^T (\text{Mask} \odot \mathbf{F} \mathbf{s} - \hat{\mathbf{y}})$$

- **End For**

- $\mathbf{s}_k = \mathbf{s}$

End For**Convergence analysis**

In this section, the convergence of the proposed approach is analyzed. In the first step, the convergence of the proposed approach is investigated for a fixed parameter σ in the following Lemma.

Lemma 1 The sequence $\{J_k = J_{HQS}^p(\mathbf{s}_k, \mathbf{z}_k)\}_{k=1}^\infty$ generated by the ISRSL0 algorithm with a fixed parameter σ converges.

Proof We calculate the following difference

$$J_k - J_{k+1} = J_{HQS}^p(\mathbf{s}_k, \mathbf{z}_k) - J_{HQS}^p(\mathbf{s}_{k+1}, \mathbf{z}_{k+1}) = \left(J_{HQS}^p(\mathbf{s}_k, \mathbf{z}_{k+1}) - J_{HQS}^p(\mathbf{s}_{k+1}, \mathbf{z}_{k+1}) \right) + \left(J_{HQS}^p(\mathbf{s}_k, \mathbf{z}_k) - J_{HQS}^p(\mathbf{s}_{k+1}, \mathbf{z}_k) \right) \quad (23)$$

Two optimization subproblems (9) conclude that two terms on the right side of the above difference are non-negative. Hence, the sequence is decreasing, i.e., $J_{HQS}^p(\mathbf{s}_{k+1}, \mathbf{z}_{k+1}) \leq J_{HQS}^p(\mathbf{s}_k, \mathbf{z}_k)$. Since all terms in the cost function J_{HQS}^p are bounded, convergence is obtained.

In the ISRSL0 algorithm, a decreasing sequence σ is used, and the denoiser coefficient $g(k)$ is directly proportional to the parameter σ . Hence, the denoiser coefficient gradually tends to 0, and we can obtain the following approximation for small σ :

$$J_{HQS}^p(\mathbf{s}_{k+1}, \mathbf{z}_{k+1}) \cong J_{HQS}^p(\mathbf{s}_{k+1}, \mathbf{s}_{k+1}) \quad (24)$$

Now, according to the definition of the ISRSL0 cost function and the auxiliary variable $\mathbf{z}$ in the HQS approach, we obtain the following inequalities.

$$J_{ISRSL0}(\mathbf{s}_k) = J_{HQS}^p(\mathbf{s}_k, \mathbf{s}_k) \underset{(9)}{\geq} J_{HQS}^p(\mathbf{s}_k, \mathbf{z}_{k+1}) \underset{(9)}{\geq} J_{HQS}^p(\mathbf{s}_{k+1}, \mathbf{z}_{k+1}) \underset{(24)}{\cong} J_{HQS}^p(\mathbf{s}_{k+1}, \mathbf{s}_{k+1}) = J_{ISRSL0}(\mathbf{s}_{k+1}) \quad (25)$$

The above result shows that the sequence $J_{ISRSL0}(\mathbf{s}_k)$ is a decreasing sequence. This result can also be obtained via Lemma 1 and (24). Since the smoothed ℓ_0 norm and the regularization term are finite, the sequence $J_{ISRSL0}(\mathbf{s}_k)$ converges.

It's important to note that this proof doesn't guarantee global convergence. Since the cost function, especially the smoothed ℓ_0 norm, is non-convex, local minimums remain challenging. Simulation results show the improvement of the proposed approach compared to the SLO algorithm.

Experimental result

In this section, we present different evaluations of the proposed ISRSLO for the application of MRI reconstruction. This section uses BM3D and Wiener filters as denoisers in the proposed algorithm. The proposed approaches are called ISRSLO-BM3D and ISRSLO-Wiener.

Data sets

During this experiment, we used two publicly available datasets, including brain and knee images, some of which are shown in Fig. 3^{18,32}.

MR images of the knee

For image reconstruction, we selected 20 samples of knee images from the Fast MRI database. Selected images from the collection "single coil val." selected from Fast MRI. Since the size of the images is 320×320 , we changed them to 256×256 for the test procedure. We have shown some test images in Fig. 3.

Brain MR images

The original data size is 256×270 , cut to reach 256×256 dimensions. We use 10 brain MR images to evaluate the proposed algorithm, some of which are shown in Fig. 3.

MRI sampling masks

The CS significantly reduces scan time for various MRI applications by minimizing the number of samples required for image reconstruction. These samples are determined through specific sampling masks, with three distinct types examined: Cartesian, Pseudo-radial, and spiral paths. The Cartesian sampling mask is the most frequently employed choice due to its robustness in data acquisition, practical implementation, and ease of reconstruction. Non-Cartesian sampling masks, such as radial and spiral trajectories, can enhance the signal-to-noise ratio (SNR) since the center of k-space sampling predominantly influences SNR. However, it's essential to note that the k-space density is non-uniform, with crucial image information concentrated in the low-frequency components situated at the center of the k-space rather than the high-frequency components at the grid's periphery.

Radial acquisition paths exhibit reduced susceptibility to motion artifacts compared to Cartesian trajectories. In our experiments, the Pseudo-radial mask is predominantly employed, given its widespread use in k-space sampling. Figure 4 illustrates this research's array of sampling masks³³.

Three evaluation criteria were employed in this study, each serving to assess different aspects of image quality. The first criterion, Peak Signal-to-Noise Ratio (PSNR), is computed as:

$$\text{PSNR} = 10 \log_{10} \left(\frac{255^2}{\text{MSE}(\mathbf{x}, \hat{\mathbf{x}})} \right) \quad (26)$$

MSE represents the mean squared error between the reconstructed image $\mathbf{x}$ and the original image $\hat{\mathbf{x}}$.

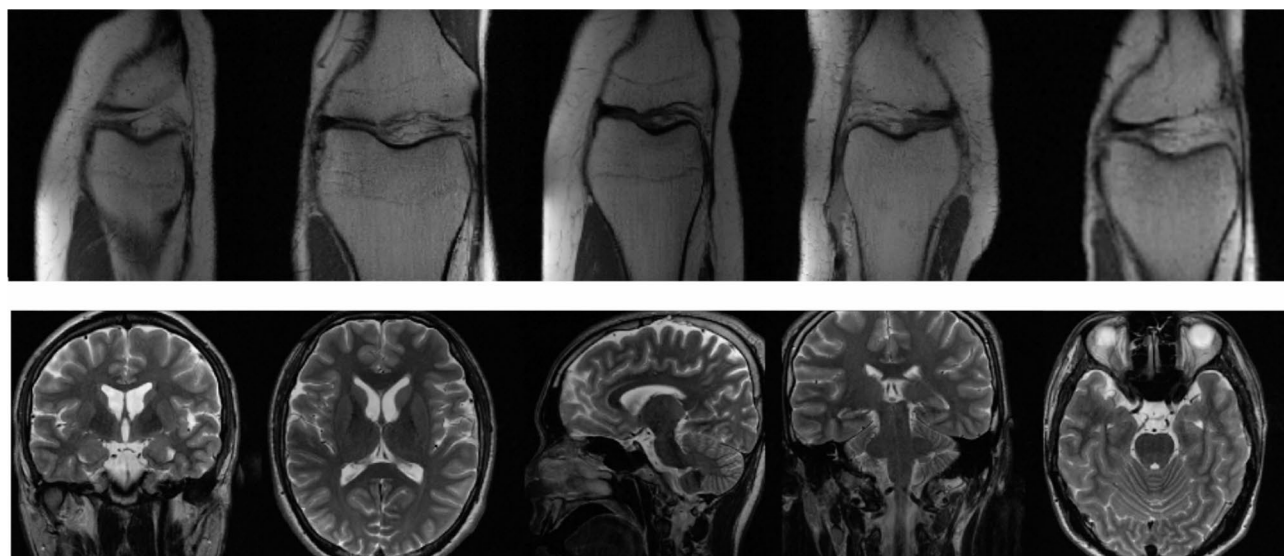


Fig. 3. Several MR images that have been used.

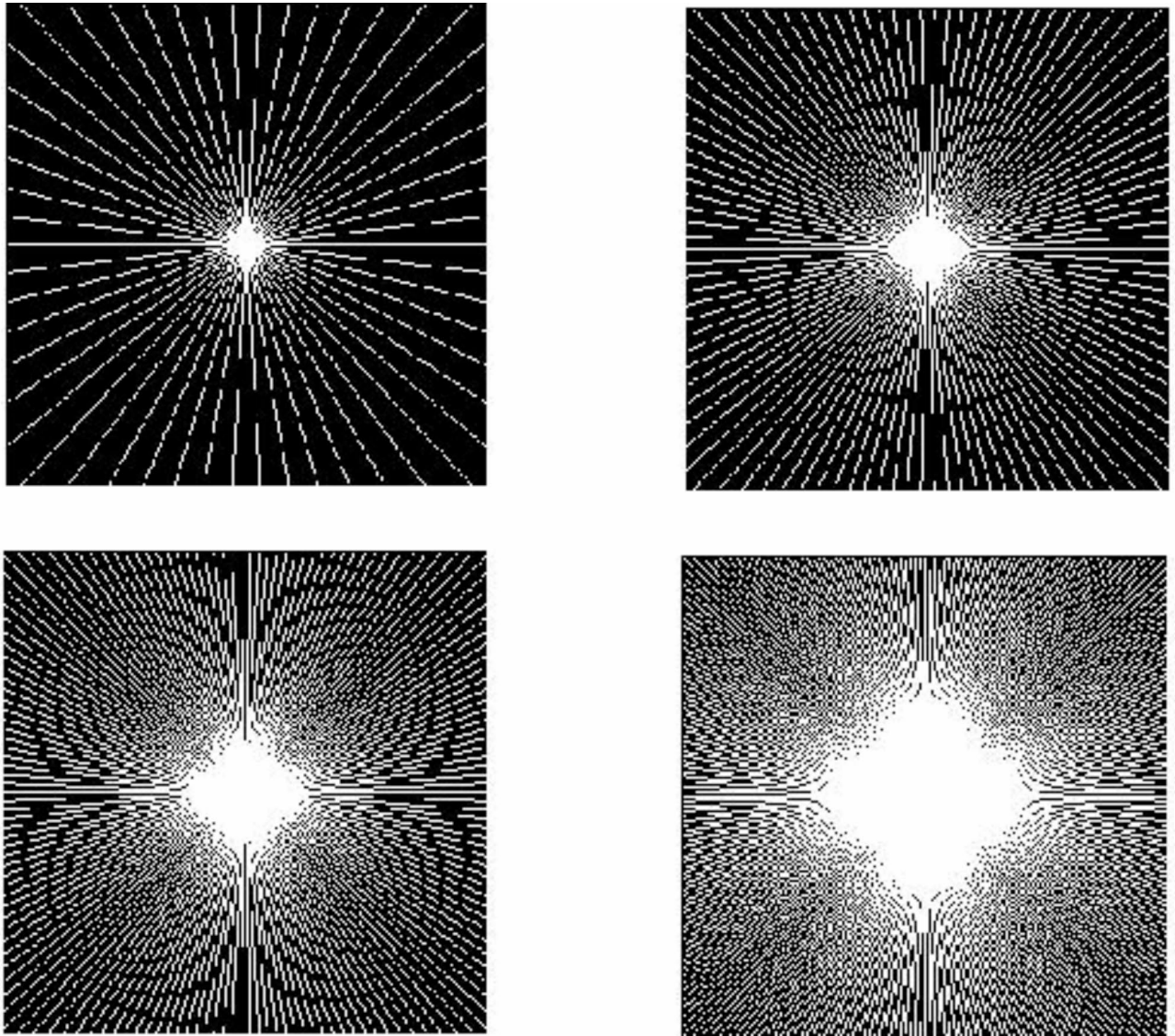


Fig. 4. Sampling patterns used in Table 2, the top row from the left is the sampling pattern with a rate of 10% and 20%, and the bottom row from the left is 30% and 50%.

The second criterion, High-Frequency Error Norm (HFEN), evaluates high-frequency image components and is determined by:

$$HFEN = H_{high}(\mathbf{x}) - H_{high}(\hat{\mathbf{x}})_2 \quad (27)$$

In this equation, H_{high} denotes the **Gaussian high-pass filter**.

The third criterion, the Structural Similarity Index (SSIM), assesses structural information preservation. It is calculated as follows:

$$SSIM = \frac{(2\mu_{\hat{\mathbf{x}}}\mu_{\mathbf{x}} + c_1)(2\sigma_{\mathbf{x}\hat{\mathbf{x}}} + c_2)}{(\mu_{\hat{\mathbf{x}}}^2 + \mu_{\mathbf{x}}^2 + c_1)(\sigma_{\mathbf{x}}^2 + \sigma_{\hat{\mathbf{x}}}^2 + c_2)} \quad (28)$$

Here, μ and σ represent the mean and variance of the signal, while C_1 and C_2 are constant coefficients. Together, these three criteria offer comprehensive insight into the quality of reconstructed images.

Comparison of SL0 with the proposed algorithm

As it was said at the beginning, the main idea of this work is to improve the classic SL0 algorithm based on smoothness regularization. By looking at Figs. 5 and 6, you can see the objective improvement of the SL0 algorithm. Also, according to the criteria compared in Tables 2 and 3, we can conclude that the ISRSL0-BM3D algorithm with different sampling rates between 2 and 3dB and the ISRSL0-Wiener algorithm between 1 and 2dB has improved compared to the SL0.

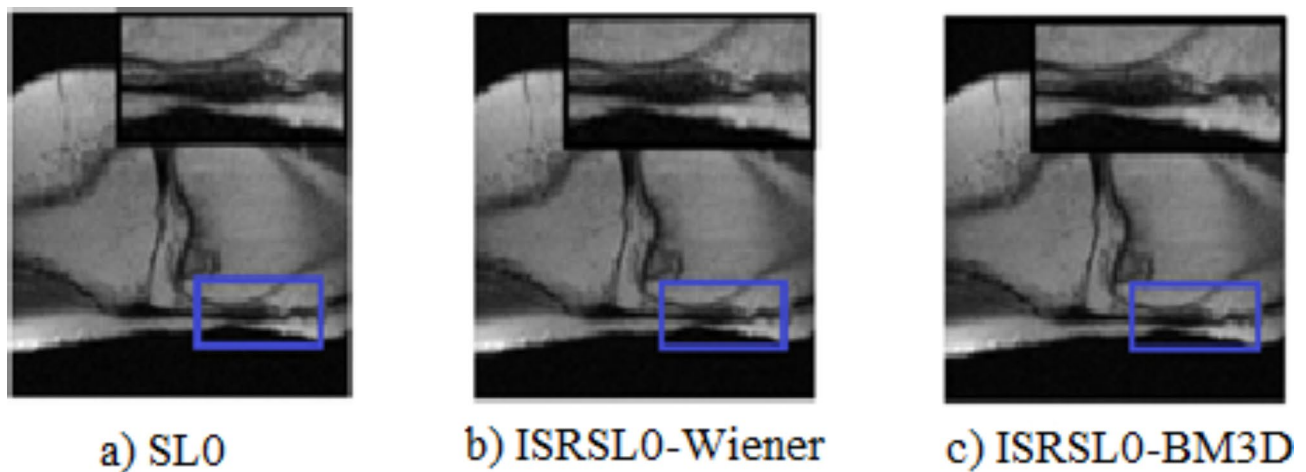


Fig. 5. Objective results of SL0 (a), ISRSLO-Wiener (b), and ISRSLO-BM3D (c) methods on knee image reconstruction with a 30% sampling rate using a pseudo-radial sampling mask.

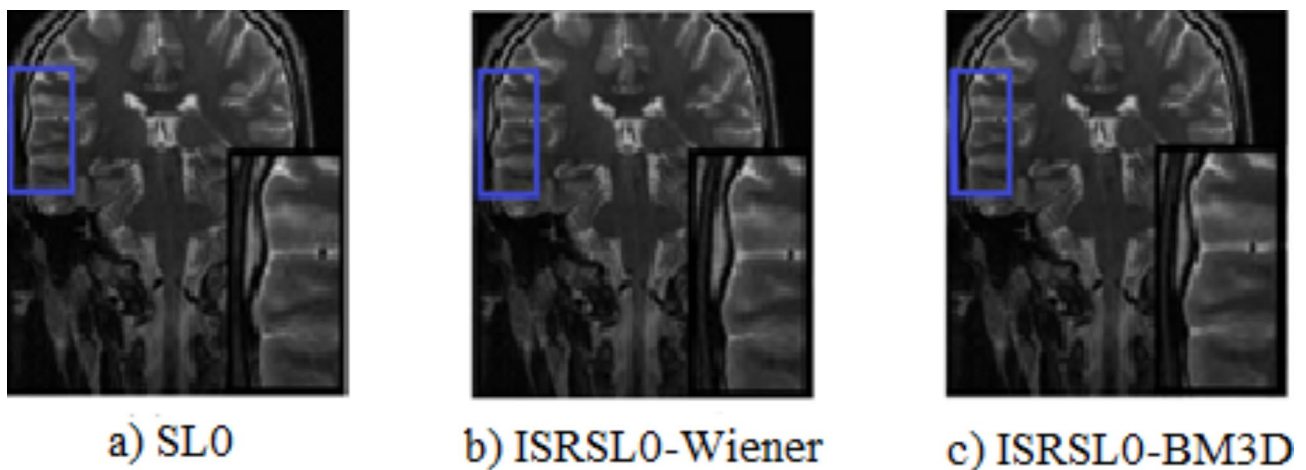


Fig. 6. Objective results of SL0 (a), ISRSLO-Wiener (b), and ISRSLO-BM3D (c) methods on brain image reconstruction with a 30% sampling rate using a pseudo-radial sampling mask.

Effect of sampling patterns

This research uses three sampling patterns with a rate of 30% to evaluate the proposed algorithm and the effect of different sampling patterns. Table 1 shows the PSNR value of different algorithms used for this test. As you can see, the **ISRSLO-BM3D** algorithm has better results than the rest of the algorithms **under all sampling patterns**. The best results for all algorithms are obtained **with a two-dimensional random pattern**.

Result of knee images

Knee data have been tested under different sampling rates with a **pseudo-radial sampling pattern**. Table 2 shows the results of the SFISTA¹³, PFISTA¹⁴, FDLCP¹², KTMR²¹, IFR-Net¹⁹, SL0²³, ISRSLO-Wiener and ISRSLO-BM3D algorithms. These results show that the ISRSLO-BM3D method provides better results, and also, the Wiener filter improves the SL0 algorithm in the case of the 10% sampling rate by more than 2dB, and for the ISRSLO-BM3D algorithm, this value reaches more than 3dB. This superiority in improvement decreases with increasing sampling rate. The objective results from the comparison of knee MR image reconstruction can be seen in Fig. 7.

Result of brain images

The brain data have been evaluated under different sampling rates, and Table 2 shows the comparison results of SFISTA, PFISTA, FDLCP, KTMR, IFR-Net, SL0, ISRSLO-Wiener and ISRSLO-BM3D algorithms, where the pseudo-radial sampling pattern is used.

According to the results of Table 3, it can be seen that the ISRSLO-BM3D algorithm performed better than all the evaluated methods. The proposed method outperforms other approaches. The PSNR improvement compared to the classical methods is obtained between 1 and 2dB under different sampling rates and compared to the

Method	Mask 10%			Mask 20%			Mask 30%			Mask 50%		
	PSNR (SD)	HFEN (SD)	SSIM (SD)	PSNR (SD)	HFEN (SD)	SSIM (SD)	PSNR (SD)	HFEN (SD)	SSIM (SD)	PSNR (SD)	HFEN (SD)	SSIM (SD)
IFR-Net	29.23 (1.68)	1.65 (0.14)	0.71 (0.13)	32.37 (2.1)	0.92 (0.11)	0.80 (0.15)	35.93 (2.41)	0.30 (0.06)	0.89 (0.17)	36.45 (2.8)	0.23 (0.04)	0.90 (0.18)
FDLCP	29.20 (1.82)	1.55 (0.15)	0.32 (0.21)	32.12 (2.5)	0.89 (0.11)	0.45 (0.16)	35.31 (2.6)	0.32 (0.07)	0.69 (0.178)	35.99 (2.73)	0.24 (0.041)	0.71 (0.2)
KTMR	29.01 (1.71)	1.7 (0.16)	0.68 (0.19)	32.15 (2.48)	0.95 (0.112)	0.79 (0.16)	35.54 (2.61)	0.32 (0.075)	0.87 (0.169)	36.10 (2.78)	0.24 (0.043)	0.89 (0.2)
PFISTA	28.03 (3.49)	1.89 (0.2)	0.31 (0.22)	31.72 (3.75)	1.07 (0.15)	0.45 (0.24)	34.88 (3.9)	0.43 (0.068)	0.63 (0.26)	36.09 (4.55)	0.26 (0.077)	0.71 (0.29)
SFISTA	28.11 (3.52)	1.8 (0.21)	0.31 (0.22)	31.74 (3.76)	1.07 (0.19)	0.45 (0.25)	34.89 (3.95)	0.43 (0.1)	0.63 (0.28)	36.10 (4.5)	0.25 (0.0.087)	0.71 (0.29)
SLO	26.22 (3.54)	2.27 (0.19)	0.55 (0.22)	30.10 (3.01)	1.5 (0.15)	0.71 (0.24)	34.75 (3.48)	0.43 (0.1)	0.86 (0.26)	35.14 (3.7)	0.32 (0.075)	0.87 (0.29)
ISRSLO-Wiener	28.37 (1.32)	1.8 (0.132)	0.68 (0.12)	31.74 (1.78)	1.03 (0.10)	0.78 (0.14)	35.66 (1.84)	0.32 (0.96)	0.89 (0.145)	36.23 (2.01)	0.26 (0.041)	0.90 (0.15)
ISRSLO-BM3D	29.30 (0.97)	1.55 (0.12)	0.73 (0.09)	32.47 (1.12)	0.87 (0.098)	0.80 (0.1)	35.97 (1.2)	0.29 (0.055)	0.89 (0.12)	36.56 (1.27)	0.24 (0.039)	0.90 (0.12)

Table 2. Comparison of the proposed algorithm with other algorithms for knee images and SD represents standard deviation.

Method	Mask 10%			Mask 20%			Mask 30%			Mask 50%		
	PSNR (SD)	HFEN (SD)	SSIM (SD)	PSNR (SD)	HFEN (SD)	SSIM (SD)	PSNR (SD)	HFEN (SD)	SSIM (SD)	PSNR (SD)	HFEN (SD)	SSIM (SD)
IFR-Net	25.60 (1.21)	2.71 (0.2)	0.67 (0.14)	31.01 (1.45)	1.28 (0.13)	0.85 (0.165)	36.48 (2.16)	0.30 (0.1)	0.94 (0.18)	37.68 (3.1)	0.23 (0.06)	0.96 (0.18)
FDLCP	25.35 (1.42)	2.8 (0.24)	0.40 (0.18)	30.9 (2.06)	1.32 (0.2)	0.64 (0.191)	36.23 (2.46)	0.31 (0.12)	0.82 (0.198)	37.10 (3.4)	0.23 (0.062)	0.83 (0.2)
KTMR	25.40 (1.4)	2.85 (0.23)	0.64 (0.17)	30.91 (2.1)	1.25 (0.181)	0.79 (0.177)	36.33 (2.40)	0.30 (0.115)	0.89 (0.0183)	37.20 (3.4)	0.24 (0.064)	0.90 (0.19)
PFISTA	24.45 (3.28)	3.20 (0.35)	0.34 (0.19)	29.05 (3.35)	1.8 (0.29)	0.55 (0.19)	34.87 (3.93)	0.56 (0.21)	0.79 (0.25)	36.69 (4.8)	0.30 (0.1)	0.83 (0.28)
SFISTA	24.50 (3.3)	3.17 (0.38)	0.35 (0.19)	29.08 (3.2)	1.8 (0.29)	0.58 (0.2)	34.89 (3.87)	0.54 (0.2)	0.8 (0.25)	36.70 (5.1)	0.29 (0.11)	0.83 (0.3)
SL0	22.96 (2.98)	3.71 (0.3)	0.48 (0.2)	26.70 (2.95)	2.55 (0.27)	0.66 (0.21)	32.98 (3.54)	0.6 (0.2)	0.89 (0.23)	34.43 (4.1)	0.46 (0.095)	0.91 (0.24)
ISRSLO-Wiener	24.70 (1.2)	3.03 (0.18)	0.62 (0.16)	29.40 (1.58)	1.67 (0.13)	0.81 (0.17)	35.65 (1.71)	0.37 (0.075)	0.94 (0.178)	36.87 (1.96)	0.3 (0.05)	0.96 (0.187)
ISRSLO-BM3D	25.70 (0.76)	2.63 (0.174)	0.67 (0.1)	31.06 (1.03)	1.19 (0.097)	0.86 (0.15)	36.81 (1.27)	0.3 (0.06)	0.95 (0.112)	37.80 (1.26)	0.23 (0.04)	0.96 (0.12)

Table 3. Comparison of the proposed algorithm with other algorithms for brain images and SD represents standard deviation.

Sampling pattern	SFISA	PFISTA	FDLCP	ISRSL0-BM3D
2D Random	35.12	35.11	35.70	36.01
Cartesian	30.20	30.20	30.54	32.61
Radial	34.89	34.88	35.31	35.97

Table 1. Average PSNR of the proposed algorithm for different sampling patterns on knee images.

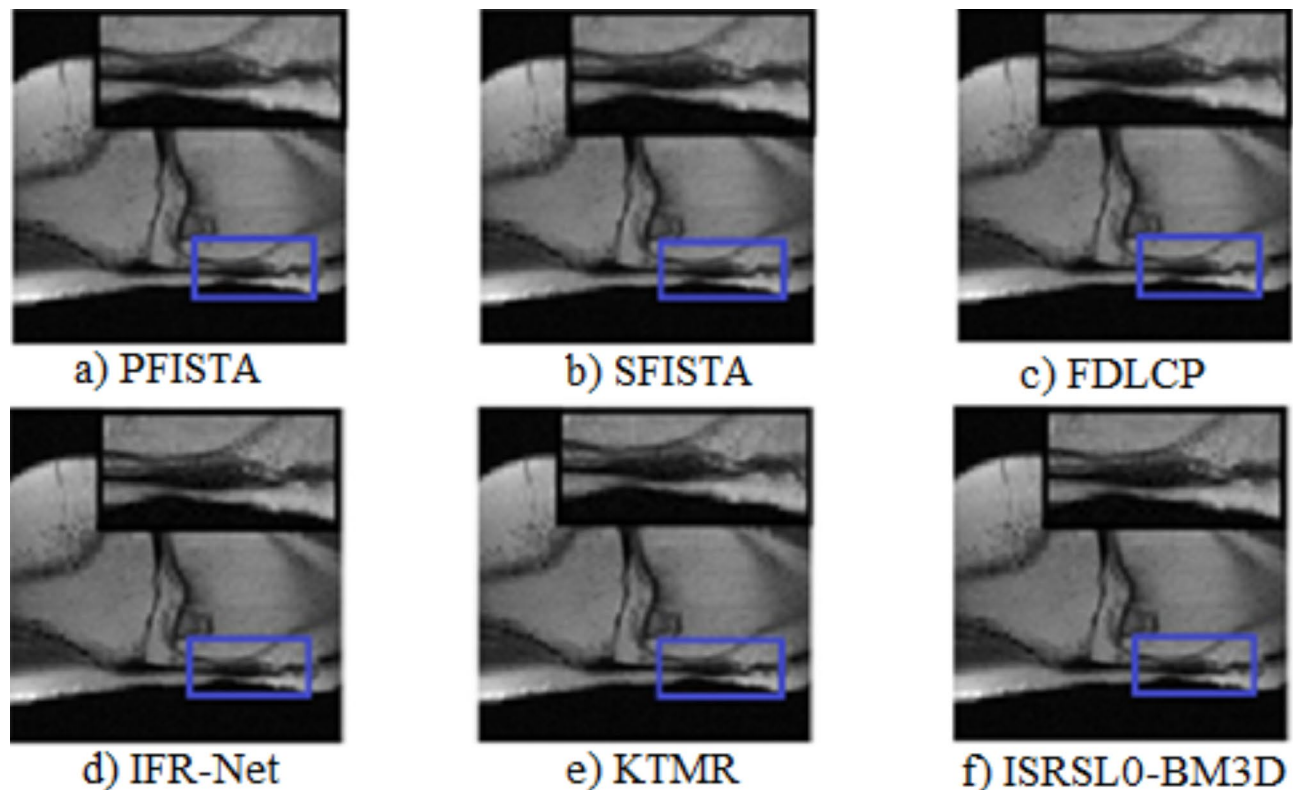


Fig. 7. objective results of different methods on knee image reconstruction with 30% sampling rate using pseudo-radial sampling mask, (a) PFISTA, (b) SFISTA (c) FDLCP, (d) IFR-Net, (e) KTMR and (f) ISRSL0-BM3D.

network-based methods between 0.1 and 0.7 dB. As stated for the knee images, this improvement decreases with increasing sampling rate. In Fig. 8, the output of the algorithms can be seen objectively.

Computation time

In this section, the computational complexity of the proposed approach is evaluated via the computation time. Note that the time measure, dependent on the skill and language programming, is inappropriate for showing computational complexity. Table 4 shows the average duration of testing different algorithms for real knee images with a sampling rate of 30%. As can be seen, the IFR-Net data-driven model has the lowest execution time.

Discussion & conclusion

This research focused on achieving the primary objective of reconstruction methods: delivering the best results with minimal complexity. To this end, we proposed a systematic process of enhancing the classic SL0 algorithm. A key element in our approach was the incorporation of image smoothness regularization, which was crucial in addressing challenges related to local minima encountered during the algorithm's execution. The proposed cost function was optimized by applying the HQS method. The proposed algorithm operates through three fundamental stages: Denoising, Sharpening, and SL0 Cost Function Optimization. Each step preserved essential features and guided the image away from local minima toward the global minimum.

We adjusted the algorithm's parameters to ensure a gradual reduction in smoothing over iterations. Simultaneously, the influence of the Sharpening was gradually diminished until only the original SL0 algorithm prevailed. Furthermore, we demonstrated the convergence of the algorithm, analytically. The results show the algorithm's success and superiority over classic reconstruction approaches. Notably, the experimental results demonstrated comparable performance to network-based algorithms, with the added advantages of simplicity

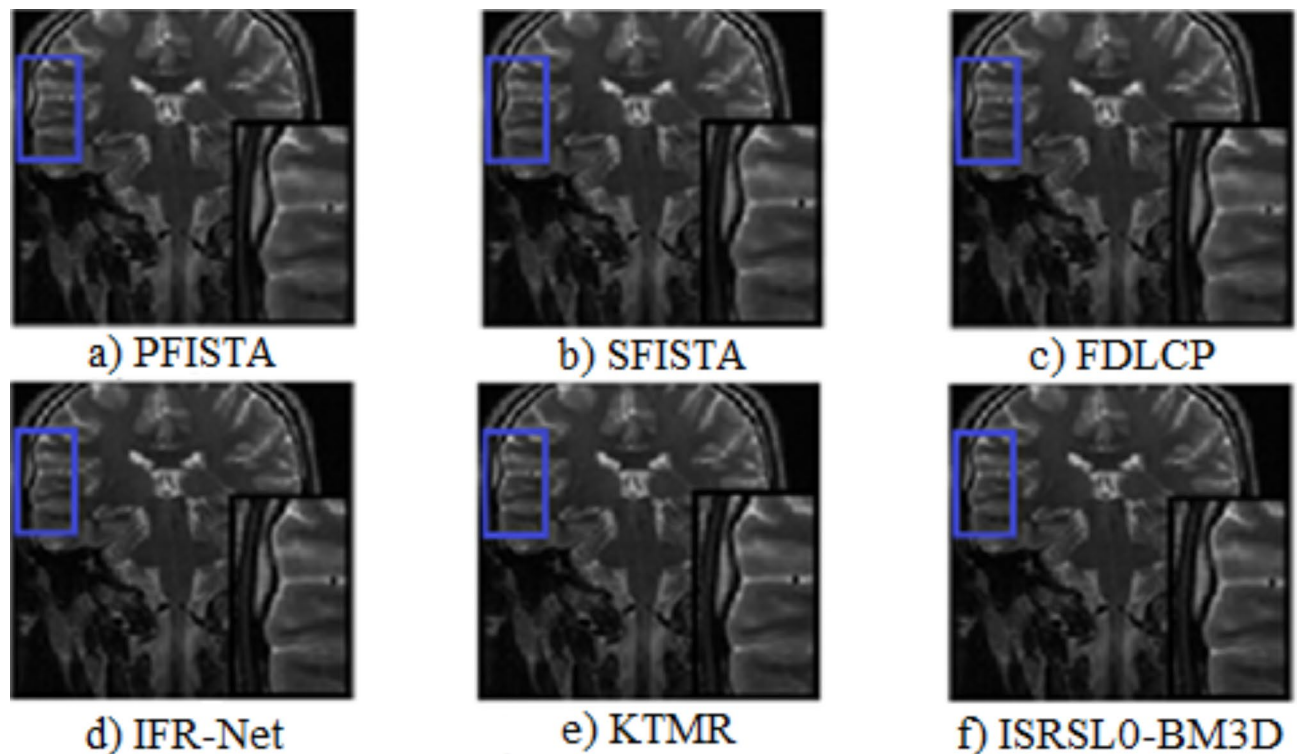


Fig. 8. objective results of different methods on brain image reconstruction with 30% sampling rate using pseudo-radial sampling mask, (a) PFISTA, (b) FDLCP, (c) FDLCP, (d) IFR-Net, (e) KTMR and (f) ISRSLO-BM3D.

Method	Computation Time (s)
KTMR	108
FDLCP	64
PFISTA	35
SFISTA	43
IFR-Net	6
SL0	7
ISRSLO-Wiener	11
ISRSLO-BM3D	106

Table 4. The results related to the execution time of image reconstruction with a sampling rate of 30%.

and low computational complexity. The results and the comprehensive evaluation against various measurement criteria indicate a marked improvement over the original SL0 algorithm. The proposed approach will be combined with the deep model to improve the reconstruction performance in future work.

Data availability

The data used in this paper are available from the corresponding author upon request.

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Author contributions

E.H.: writing original draft, review and editing, software, methodology, formal analysis, validation. A.G.: review and editing, formal analysis, methodology, conceptualization, investigation, project administration, supervision.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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